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Review Article

The mechanisms of forensic vacuum metal deposition (fVMD): A literature review

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ABSTRACT

Since the early 1970s, forensic vacuum metal deposition (fVMD) has been used in operational casework. It is renowned for its sensitivity and ability to work on aged impressions and difficult substrates. However, the precise mechanism by which this development occurs remains unclear. This article examines the current literature on fVMD to see what is known about the technique, what is unclear, and what are the next steps that should be taken. This review shows that, while the operational utility of the technique is not in question, none of the currently posited theories can explain all the observed phenomena. They are therefore, at best, incomplete, and further research is required.

1. Introduction

One of the primary goals of forensic casework is the association of a suspect individual to a scene. The best way to do this remains the use of fingermarks, which can not only serve to identify an individual, but the placement, distribution, and manner in which the fingermarks were deposited can furnish critical contextual information to the investigation. Of course, to be able to interpret the fingermarks, they first need to be located, documented, and collected. While forensic specialists possess many techniques to develop said fingermarks, one of the most sensitive tools in the arsenal is forensic vacuum metal deposition (fVMD).[1] Often referred to simply as VMD, fVMD is a category A process [1] used daily by police forces around the world to develop fingermarks in operational case work. Since its development in the 1970s, fVMD has been recognized for its high sensitivity and versatility, especially for cold cases (historical files wherein all the previous investigative leads had been exhausted) [1–5]. Some examples include the *Last Call Killer* case [2,6], a 20 + year-old impression developed by the Royal Canadian Mounted Police in Surrey, British Columbia [7] and a 41 year-old impression developed by the Royal Canadian Mounted Police in Montreal, Quebec [8]. fVMD is also known for working effectively on substrates where other techniques tend to fare poorly. Some examples include textiles [9,10], polymer bank notes [11–13], exhibits that had been wetted or exposed to the elements [3,14,15], and surfaces exposed to high temperatures [16,17], including cartridge cases [18–20].

Despite this resume the technique remains unrefined as the precise mechanism of development remains unclear. While there is a consensus within the forensic community, there is no conclusive evidence. As an example, *Fingermark Visualisation Manual,* 5.VMD.17 refers only to a "general agreement on the theory associated with normal development of prints" [1].

The objective of the current paper is therefore to perform a review of

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[1] Author's note: The technique being discussed is commonly referred to simply as vacuum metal deposition (VMD). While VMD is frequently used in many fields, including manufacturing, its utilization in forensics is a highly specific, non-standard use of the technique that does not easily translate to other fields. This paper therefore uses the less common, but more precise, fVMD to ensure the information is properly contextualized.

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the current knowledge surrounding fVMD to assess the current understanding of the deposition mechanism. This would provide a theoretical basis from which future fundamental research projects into this question can be premised.

2. History

2.1. Initial discovery (1964–1968)

On November 13th, 1964, Dr. Samuel Tolansky, a professor with the University of London physics department (London, England) specializing in interferometry, wrote to the British Home Office regarding a nuisance issue that sometimes occurred while depositing silver films onto glass to create optical surfaces for his instruments. Occasionally a fingermark would be developed on the glass substrate, ruining the homogeneity of the silver (Ag) film deposition. While this was problematic for interferometry, he nonetheless reasoned "that although the developed up prints were a nuisance to him in his work on interferometry they might be of use to someone else: the police force for example" [21]. The letter proposed investigating this for forensic purposes, including as a proof a "extremely faint", otherwise undetectable fingerprint developed using a "(deposited) layer of silver about 250 Å *(25 nm)* thick" [22]. However, as the proposed method was on glass. As the suggested development techniques of the time (fingerprint powder) already worked well on glass, the proposal was not pursued [4]. The situation remained unchanged until 1968 when a group of workers out of the Préfecture de Police in Paris, led by P. Theys, reported similar findings whereby a volatilized metal mixture was able develop fingerprints on paper [23]. Not only were fingerprints developed on paper, a problematic substrate, but the metal was able to develop fingerprints aged out to 11 months. This was sufficient to renew Tolansky's interest in the technique, who initiated a research project to be pursued by a PhD student, David Sutton Hambley [21].

The first step involved examining the data and process of Theys *et al.*, including why a volatilised metal mixture was used. The reported technique involved placing the sample in an evaporator, reducing the vacuum to 10^{-5} torr, holding it at that level for 50–60 min, then gradually increasing the temperature of the mixture until volatilization occurs, at which point it was held until the fingerprint was deemed sufficiently developed. The mixture reported by Theys *et al.* was Zinc: Antimony: Copper. The precise ratio is unclear, as Theys *et al.* reported a ratio of 73: 21.5: 5.5 respectively, though Hambley reported a mixture of 56: 16: 4 for subsequent samples. The reason for the disparity is unknown. However, it proved to be a moot point, as an analysis of a sample prepared by the French team but analyzed by Hambley using an electron probe microanalyzer showed only zinc (Zn) was found to have been deposited on the surface. Upon further discussions with the French team, it was determined the initial (re)discovery had occurred accidentally while preparing a sample for an electron microscope, wherein they were evaporating impure chrome to coat a surface. It was on this information that the subsequent tests were based.

It was from these independent, yet associated, discoveries that fVMD was derived. It is important to note at this point in the discussion that, as both discoveries were serendipitous, any initial understanding was phenomenological rather foundational. While this is not a problem *per se* as some of the most significant scientific breakthroughs started with phenomenological studies, this realisation will help the reader understand some of the issues the initial researchers had in examining this question, detailed further below.

2.2. Initial research (1969–1976)

The first attempt to understand this process was undertaken by Hambley in concert with a group of scientists (Stroud, Collins & Coles) from the Atomic Weapon Research Establishment (AWRE) in Aldermaston, Great Britain, and a group of scientists from the Home Office Scientific Research and Development Branch (HOSRDB), London. These efforts still serve as the basis of the current understanding of the technique. From this research came the following essential insights:

1) High quality fingerprints could be developed on paper (a major improvement for the time) and cloth [21, 24–26]
2) Zn was found to be the only metal deposited on the samples from the French team, however when tested alone, Zn deposited as a single metal generates a spotty deposition [21].
3) The spotty Zn deposition could be rendered uniform by first seeding the surface with gold (Au) [21]. Of note is the minute quantity of Au that was determined by extrapolation to give the best results; a film of 5Å (0.5 nm), determined by Hambley [21] versus a film of 3Å (0.3 nm), determined by Collins [26]. Though referred to as a film throughout the papers, in both cases it was recognized that the films were likely discontinuous. It was noted, however, the optimum deposition conditions (including the quantity of Au) are heavily substrate dependent [26].
4) When performing a sequential deposition starting with Au, cadmium (Cd) generated the best contrast, with Zn a close second [21].
5) Single metals can also be used, and when a large amount of material is deposited, generates a colour change. From Hambley, an exposure time sufficient to generate an Au film of 800Å (80 nm) resulted in "a general blue-grey colouration" on the paper and "pink-magenta regions" in the areas with the matrix. Meanwhile a silver layer "some 500 Å *(50 nm)* thick" generated a light brown colour [21]. No explanation for the colour shift was given.
6) That only seven compounds of the 50 that were tested appeared to inhibit Zn deposition: hexadecenoic acid, octadecanoic acid, cholesteryl oleate, glycerol trioleate, arginine monohydrochloride, leucine and threonine [21]. Given six of the seven (all except leucine) are hydrophobic, it was concluded the inhibition was caused by the lipid components of the fingerprint matrix. This was further validated by Collins and Cole [24–26], Thomas and Reynoldson [27].
7) Given the hydrophobic nature of the targeted compounds (see point 6 above), it was theorized that fVMD would still work on fingermarks that had been exposed to humidity. This proved to be correct, with fVMD showing no degradation after 144 h of exposure at 90 % humidity [21]. This was further validated by Collins and Coles [24–26].
8) fVMD was particularly effective on aged impressions. Theys *et al.* had demonstrated a longevity out to 11 months [23], Hambley was able to demonstrate development on paper even after 2 years [21].
9) fVMD is sensitive down to a monolayer of lipid contaminant, the equivalent of a lipid monolayer [27,28].
10) fVMD is extremely sensitive to the type of substrate. Seemingly identical substrates could generate normal development, reverse development (where the Cd deposits primarily on the ridges) [29], or fail to develop at all [21, 30–32].

This research also generated the first proposals for a mechanism of development of fVMD. Stroud, working with Hambley, also proposed a theory based on the differential adsorption of the metal to the surface [33,34]. However it was found to be lacking [26]. This theory will henceforth be referred to as the condensation theory. The second theory came from Collins, Cole, and Stroud [26]: "It may be that during the gold evaporation the deposited gold becomes intermixed with the grease and print remains essentially as a surface grease pattern, which inhibits the subsequent deposition of cadmium." [26]. This will henceforth be referred to as the absorption theory. Both theories will be further discussed in depth below.

As stated above, despite these insights being over 50 years old, they

still form the basis of our understanding of fVMD. The sequential gold-zinc deposition (AuZn) is still considered the most effective fVMD option for general use [1,4,9,11,13,35-37] and the absorption theory is still *de facto* explanation in the literature [1,4,38,39].

2.3. Initial introduction to operational application (1975–1982)

With the initial demonstration of the theoretical validity and applicability of fVMD, by the mid-1970s the research had turned towards operational use, spurred on in part by a spate of IRA bombings and the need to develop fingerprints on polyethylene (then referred to as polythene) [5]. At the time the fingerprint development on polyethylene (PE) was done using fingerprint powders, but the powders only worked on fresh impressions [40].

The first operational introduction was via a technical memorandum by Kent, Thomas and East [40] (HOSRDB) examining fVMD on PE using a sequential deposition of Au (or Ag) followed by Cd. It examined both laboratory impressions using natural prints from multiple donors on fingermarks ranging from 1 h old to 28 days old and stored under various conditions as well as actual casework. The results were excellent, with 50–75 % of the laboratory impressions being deemed usable for identification even after 28-days. Based on those results, they then applied fVMD to operational exhibits and generated 26 potentially usable impressions on 361 exhibits from 97 cases over a period of 4 months. Of note for this paper is the authors' comments on the occurrence of reverse development, that the Cd appeared to preferentially deposit on the fingerprint matrix rather than the substrate. They stated they believe this to be due to "the absorption by the polythene of the mobile organic constituents of the deposits leaving a more solid residue of inorganic material … penetration of the gold may then occur to a greater extent in the polythene" [40]. However, in a later report, it was also noted that "During (the earlier) trials it was noticed that on some articles it was not possible to develop fingerprints because metal could not be deposited anywhere on the surface" [30].

By 1982, the operational use of fVMD had evolved to the point where the HOSRDB published a user guide for operational use as recommended technique for the development of fingerprints on PE. The guide included aspects such as the treatment process, exhibit handling and integration with other development techniques, but also officially replaced sequential gold-cadmium deposition with the AuZn sequence, for health and safety reasons. In addition, it clarified that deposition times are dependent on the substrate, that it can be used on most uncontaminated surfaces, on aged impressions, and that while it can be used on fabric, no usable impressions had yet been developed in operational files. Also of interest for this paper is the observation that the Zn coating "may be oxidised, and become transparent" [41]. We can therefore see fVMD is already becoming established as a sensitive technique that works on difficult and aged samples.

2.4. Continued refinement (1982 – present)

The research over the next several decades further cemented that reputation. This included continued research into fVMD on various polymers. Pounds showed fVMD developed more impressions than powder suspension on PE [42]. D.S. Smith showed fVMD worked successfully on polyvinylchloride (PVC) [43]. Reynoldson and Reed compared fVMD using AuZn and small particle reagent (SPR, a.k.a molybdenum disulfide powder) on PE and found fVMD was easier to use and developed twice as many impressions [44]. As Kent [41] before though, they did also note "fingerprints once developed tend to fade due to oxidation of the thin metal film." Misner, situated across the Atlantic with the Royal Canadian Mounted Police, found fVMD surpassed the relatively novel cyanoacrylate fuming (CA) technique on PE, even when the CA was subsequently treated with fluorescent dye stain [45,46]. Comparisons by Kent, Gillet and Lee against fingerprint powdering using aluminium or lead on PE again showed fVMD using AuZn to be superior [47]. Shipman showed fVMD outclassed CA and fluorescent dye when tested on a series of operationally relevant forensic materials, including cartridge cases, plastics, gloves, and handguns. Of note in this case was the use of an artificial matrix to control for donor variability [20]. Similarly, a comparison by Masters showed fVMD using AuZn to be superior to fingerprint powdering using black powder, even when humidity is added to the fingerprint to increase the adherence of the powder. Of particular interest for this paper however is the fact Masters witnessed reverse development in some cases, despite using a glass substrate [48]. G. Smith also noted reverse development and attributed it to "epitaxial growth" of the zinc on sodium chloride crystals [49].

Also due to reverse development, Grant, Springer and Ziv examined PE as a problem substrate, examining whether the level of a lubricant amide additive in PE bags could explain the variability of Zn deposition. They found the presence of the lubricant amide non-significant in the type of development, but that the type of PE was significant. Low density PE (LDPE) had a glossy, smoother surface compared to high density PE (HDPE) and had a greater tendency to generate reverse development [50]. Ziv and Springer also demonstrated, both operationally and in the lab, that washing contaminated PE exhibits prior to fVMD treatment could be beneficial, but could also have a negative impact [15]. Morris *et al.* showed that fVMD can have operational benefits beyond fingerprints, by using it to assist in examining credit cards [51]. Batey *et al.* showed the benefit of using fVMD in several operational cases, all on plastic, included an exhibit that was at least six years-old and another where the exhibit has been in a river for almost a day prior to treatment. Of note, fVMD was specifically cited as being the technique of choice for "old or difficult surfaces" [3]. In an interesting application to polymer exhibits, Meikle *et al.* found treatment with fVMD did not interfere with isotope ratio mass spectrometry analysis of the polymer substrate [52].

In a callback to fVMD's roots, Misner found fVMD to be more effective than DFO on paper initially, but over time found the matrix absorbs into the paper, at which point DFO becomes more effective [46]. This is in comparison to Godsell, Vincent and Lloyd who found fVMD with AuCd surpassed ninhydrin in quality of development, albeit when the samples were stored in a humid environment [53]. In a more modern variant, Sherriffs *et al.* found fVMD using AuZn or AgZn surpassed Lumicyano™ when attempting to develop natural matrix impressions on thermal paper. The best sequence (of AuZn vs AgZn) depended on the type of thermal paper being treated [54]. Similarly, Wang *et al.* found fVMD (AuZn) was surpassed by sequential 1,2-indanedione and ninhydrin, except when the bills had first been exposed to water [55]. Suzuki *et al.* were able to demonstrate fVMD was even effective when the paper was treated with ferromagnetic ink (in the form of train tickets), as well as on foam styrol packaging and PE. They were also able to further demonstrate that fVMD is effective on aged impressions, as one of the impressions developed on the tickets was over 4 years old (as confirmed by the date on the ticket) [56].

In another callback to fVMD's roots, Albinson showed that only fVMD (using AuZn) and radioactive sulfur dioxide were capable of developing impressions on fabric, assuming the fabric was clean and had a density of at least 3 threads/mm [57]. This requirement for a dense material was echoed by Fraser *et al.*, who found fVMD using AuZn could develop fingermarks on fabrics, but the quality of the impression was highly dependent the density and hydrophobicity of the tested fabrics [58–60]. Follow-up work by Knighting *et al.* using Ag-only deposition on dark fabrics generated usable ridge detail on polyester, but only empty impressions (voids that show evidence of handling, but no ridge details) on the other fabrics [61]. Horvath took this a step further and examined AuZn on fabrics as well as a whole series of alternative metals (metals other than sequential AuZn), including copper, sterling silver, aluminum in various combinations [10]. Horvath found a sequential deposition of copper-Zn gave the best result, followed closely by a sequential deposition of aluminium-Zn. But again, the results were highly variable, dependent both on substrate and donor. Proving how variable the substrates themselves can be, Woodcock *et al.* tested fingerprint

development on pangolin scales and rhino horns and showed the combined sequence of fVMD (AuZn) with CA was the most effective treatment.

While multiple different metals were examined in the initial research, most of the research focused on AuZn. But, as Horvath showed above, alternative metals have their uses, particularly on substrates ill-suited to AuZn. Brewer examined a series of alternative metals on fired cartridges (one of the most difficult substrates upon which to develop impressions) and found Ag deposition to be the most effective treatment for brass cartridges, with AuZn the most effective on the plastic of shotgun shells [18,19]. Christofidis *et al.* reinforced how difficult a substrate fired materials are when they compared AuZn vs sequential silver-zinc deposition (AgZn) using natural matrix on fired brass cartridges and showed only 1 impression out of 20 was potentially usable for identification [62]. Chelli showed alternative metals could be cheap and effective on most surfaces, including on duct tape where AuZn fares notoriously poorly [63]. Gunarante, Knaggs and Stansbury compared AuZn to aluminium (Al) on natural impressions on a series of consumer plastics, finding Al fared comparatively better on fresh prints, but comparatively poorly on aged impressions [64]. In an interesting twist, Yu found thermally depositing Zn oxide (ZnO) on natural impressions deposited on polyethylene terephthalate (PET) allowed Zn deposition without Au seeding [65]. Similarly Tiwari *et al.* also examined sequential Au development when it was followed by either ZnO, Zn sulfide (ZnS) and Zn on rough aluminium, with AuZnS providing the best contrast [66].

In terms of interest for this paper however, the most impactful examination of alternative metals was done by Philipson & Bleay, when they examined alternative metal development (specifically copper, Ag, Au, tin and indium, with some sequential deposition) on problematic substrates (including HDPE, LDPE, PVC and seven others) [67]. Ag was found to give the best results on surfaces where AuZn development was poor, and sequential Au-Zn-Ag treatment was able to develop additional impressions than AuZn alone in approximately 10 % of cases. In cases of empty prints, subsequent Ag deposition often generated a usable impression. But the real strength of the paper was in the in-depth analysis. They, as Hambley [21] and Collins [26] noted, single-metal deposition tended to result in a visible contrast and colour change between the matrix and the substrate. In the case of Ag, as an example, it generated a "brownish pink background" with "yellow ridges". For Au, "blue background" with "pink ridges". They also tested surface tension as a factor in the differential deposition of AuZn, using surface tension test pens. They found "there was no consistent trend between surface tension and case of development using gold and zinc." They did present a potential mechanism of development by single metals, whereby the contrast is attributed to differences in the metal nuclei size and density between the matrix and the substrate. This was confirmed with atomic force microscopy (AFM), which found that Ag deposited on uncoated PET showed small, tightly clustered nuclei on the PET and larger, more disperse nuclei on the matrix (see Fig. 1). This mechanism was summarized schematically in the figure below (see Fig. 2) and is cited in the Fingermark Visualisation Manual as the method of contrast development for single metal deposition [1,5]. Note the figures shows the nuclei as having been absorbed into the matrix. This can therefore be considered as an application of the previously mentioned 'absorption theory'. The only mention of a mechanism for the colour shift in the metals, beyond the observation of the size and density, is being "differences in nuclei size and density and consequently the colors observed transmitted light" [67].

Gaskell *et al.* [68] also used Ag fVMD when testing the effect of surface contaminants on various forensic development techniques. Ag fVMD proved to be the best technique on oily impressions on uncontaminated surfaces. However, it was noted to be "ineffective" when the surface had a pre-existing layer of any of the oily contaminants. This seems to confirm both the sebaceous materials as being the cause of the

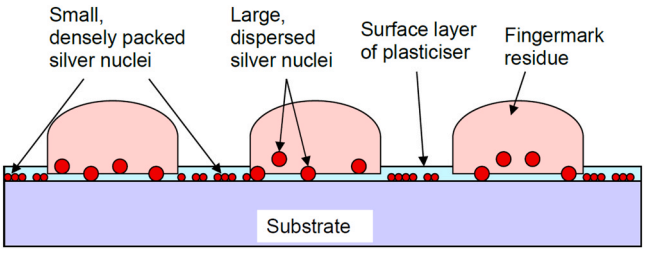


Fig. 2. Schematic diagram of silver vacuum metal deposition on a plasticised surface. Schematic diagram of silver vacuum metal deposition on a plasticised surface showing the differential deposition on the matrix versus the substrate [5], pg. 3.VMD.7.

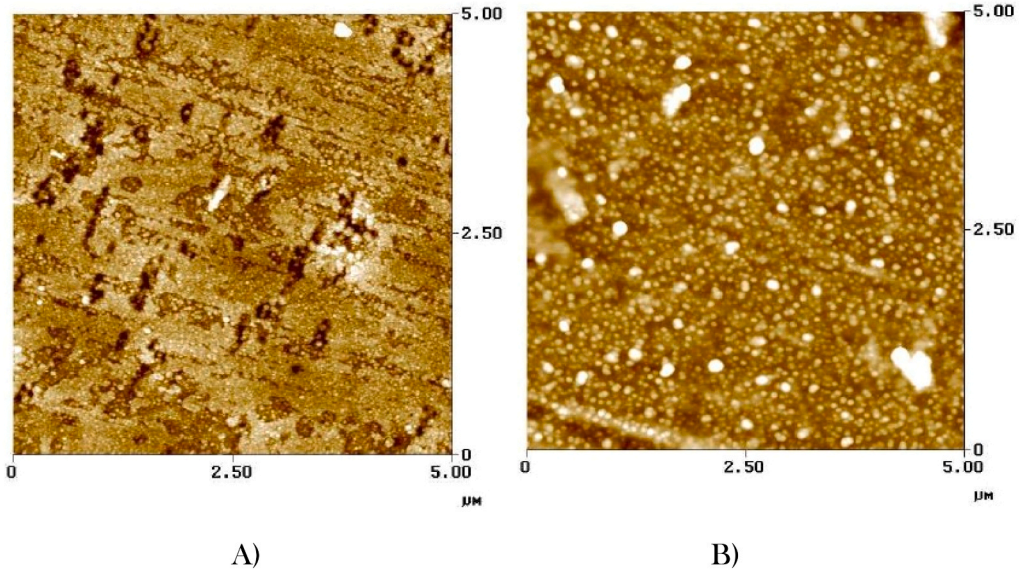


Fig. 1. Atomic Force Microscopy image of silver nuclei after vacuum metal deposition. AFM images of an unplasticized polyvinyl chloride bag after vacuum metal deposition using silver. A) AFM imaging of the background. B) AFM imaging of the ridge region. The nuclei (light dots) are smaller and more tightly packed on the substrate than on the ridge (Philipson and Bleay [67], as cited in [5], pg. 3.VMD.8).

differential deposition and the sensitivity to surface effects.

Charlton *et al.* [90] examined Ag and AuZn deposition on glass, polyvinyl chloride, and polyethylene terephthalate and confirmed can Ag created a contrast even when AuZn could not. They also confirmed time of flight-secondary ion mass spectrometry could effectively develop impressions following fVMD treatment.

Another interesting set of observations using Ag comes from Dominick, Nic Daeid and Bleay. While examining the effect of high temperatures on nonporous surfaces, they noted fVMD using Ag deposition was able to develop impressions even after having been exposed to temperatures of over 900°C for 20 min [16]. This echoes previous comments by Bleay, Bradshaw and Moore when they noted blood deposits exposed to high temperatures results in differential surface oxidation, "(a)s a consequence, processes sensitive to surface condition such as VMD are capable of detecting fingerprints in blood have been even after to exposure to temperatures of 900°C" [17]. These seem to indicate sebaceous materials are not the only possible target. The potential implications of these observations for the development mechanism will be further discussed below.

A final observation regarding alternative metals come from Steiner & Bécue, when they examined the effect of water immersion on subsequent fVMD treatment using AuZn, Ag and sterling Ag [14]. The fVMD still worked very well, particularly on PVC. However, there were some drastic colour shifts away from the norm. Fig. 3 below shows and example of unwetted sterling Ag vs. post-wetting sterling Ag [14]. The resistance of fVMD to wetted materials was also tested by Nic Daeid, Carter and Laing, who showed fVMD using AuZn narrowly performed better than powder suspension on wetted plastics, generating more and better quality impressions overall but performing poorly on plastic cowlings [69]. Both tests further supporting the idea the fVMD interacts with the sebaceous (i.e. hydrophobic) components of the matrix. A more direct test of this was done by Pollitt *et al.*, who examined fVMD on fired ballistic material (brass cartridge cases) using AuZn and AgZn. Not only was fVMD able to develop some usable details, but an associated GC-MS analysis of the impressions showed that the quality of the results were directly correlated with the quantity of stearic acid, palmitic acid and squalene in the matrix [70]. This confirms the observations made by Hambley [21].



Fig. 3. – Dry impression versus wetted impression after vacuum metal development with sterling silver. The dry half of the impression (left) versus the wetted half of the impression (1 h immersion, right), developed using sterling silver on polyethylene. Image source 5b from Steiner and Bécue [14].

A further evaluation of the matrix effects came from Popov, Sears and Jones who performed atomic force microscopy (AFM) on an impression that had been deposited on silicon and found the edges diffused up to 4 μm before regressing [71]. Bright *et al.* examined the effect of vacuum on the fingerprint matrix and found drastic reduction, equivalent to 5 weeks of aging, in the concentration of several chemical compounds targeted by fVMD, including stearic and palmitic acid [72]. When Jones, Downham and Sears did a nanoscale analysis of the fVMD AuZn treated impression following a pre-treatment by CA, they found that CA reacts primarily with the eccrine secretions from the pores, while the fVMD deposits both on the CA polymers and the ridges between the pores [36].

Finally, the author will note the experiments of Edwards [73], Kent & Stoilovic [74] and Turner *et al.* [75], who demonstrated that sputter deposition can be as effective as the standard thermal deposition systems, with the noted advantage of having a greater control over the energy of deposition.

2.5. Polymer banknotes (1999 – present)

A lot of the current interest in fVMD was due to the introduction of polymer banknotes. Banknotes have long been a difficult substrate. As far back as 1975, Newton found fVMD using AuCd an effective method to develop fingerprints on bank notes, those she notes old or well used notes did not give good results [76]. However, fVMD was often found to be the only technique for developing impressions on polymer banknotes. This result was echoed by Flynn, Stoilovic and Lennard who found the fVMD treatment after CA surpassed fingerprint powders, iodine-benzoflavone, CA alone (with or without dye stains), 4-dimethylaminocinnamaldehyde and physical developer, and was the only technique able to develop impressions older than a week [12]. Davis *et al.* found UK polymer banknotes treated with elemental copper and visualized using an infrared light produced usable impressions, especially if the impression were subsequently lifted with a gelatin lifter and treated with rubeanic acid [77]. Furthermore, the current interest in fVMD by the Royal Canadian Mounted Police was spurred by the conversion of the Canadian currency to polymer banknotes, primarily due to the fact fVMD using sequential CA - AuZn treatment generated the best results [7,11,78].

The treatment of banknotes was also at the root of the research by Jones *et al.*, when investigating the development of impressions on the new Australian polymer banknotes [13,79]. Manufacture from a polypropylene base and including a final protective overcoat, it was determined that treating it soon with CA, followed quickly by fVMD was the best development technique, albeit the efficacy reduced as the banknote became more worn [13]. But of more importance for this paper, was the subsequent in-depth studies performed by Jones *et al.* on normal and reverse development on PE [80]as well as the effect of substrate, donors and age on fVMD development [81]. In the first paper, Jones *et al.* [80] measured the amount of gold deposited onto various polymer substrates (with impressions) using a quartz crystal monitor and evaluating the impact on subsequent zinc deposition using densitometry. They found the plastics could easily be differentiated by the resulting development. For the first group, determined to be LDPE, the development progressed from normal (standard) development, to reverse development, to no contrast development as the quantity of gold deposited increased (prior to the zinc being deposited). For the second group, determined to be HDPE, the development progressed from normal development, directly to over-development, then to a poor-quality normal development as the

quantity of gold deposited increased (prior to the zinc being deposited). Subsequent inductively coupled plasma-mass spectrometry (ICP-MS) and x-ray fluorescence (XRF) measurements measuring the total amount of each metal deposited per square cm showed no significant difference in the density of gold deposited. However, there was a significant difference in the amount of zinc deposited. The derived conclusion was it was the morphology of the deposited gold which affected the deposition of the zinc. It was proposed that, on LDPE, following continued deposition the gold clusters grow in size rather than in number and eventually reach a point where the zinc will no longer adsorb. In comparison, the HDPE forms smaller, more densely packed clusters, which grow in number instead of size during continued deposition of gold. From Jones [80], pg. 85 on HDPE plastics "As more gold is deposited, the number of small clusters is increased rather than the size of the clusters. Since the gold clusters remain small, they do not reach the critical morphology that prevents zinc nucleation." This will henceforth be referred to as the 'morphology theory' and will be further discussed below.

A follow-up paper by Jones *et al.* [81] further delved into the effect of polymer type on fVMD. Again, using a quartz crystal monitor, densitometer and ICP-MS, the paper once again confirmed the sensitivity of fVMD to substrate effects. Each of the polymers tested, (HDPE, LDPE, PE, PVC and polypropylene (PP)) required different gold counts for optimal development, to the point the authors suggested polymer type be determined prior to development. The authors also noted variability within the same types of polymers, attributed to different additives, but noted the variability within polymer types was lower than between different polymer types. Empty prints also proved problematic, in particular on the PE and PVC substrates with fresh, heavy (i.e. a large amount of matrix) impressions. The posited theory for the empty prints was the matrix was seeping into the valleys, and the reason this occurred on PVC and PE was lower adhesion between the matrix and the polymer. Of note for this paper was an observation regarding the PP. The impressions would develop as though they were reverse impressions, but even the valleys would fill in. As a result, the zinc appeared to have deposited on the impression, including the valleys, but nowhere else on the substrate. The authors note "The (PP) results do not fit with the present understanding of VMD development" ([81], pg. 174).

Not covered in the above-mentioned papers, but covered in Jones [79], was an X-ray photoelectron spectroscopy (XPS) analysis of the gold-zinc deposition, where it was found zinc did deposit on the gold clusters even if the layer was not visible. Upon exposure to air, it was proposed this zinc layer oxidizes and combines with atmospheric components to deactivate the gold clusters, allowing for re-treatment.

This re-sensitization of the surface once again confirms the sensitivity of fVMD to the surface effects. It also supports the idea that it is the physical characteristics of the cluster that change. This was further buttressed in a follow-up study by Dai *et al.* who analyzed fVMD gold depositing on a polyvinyl film using transmission electron microscopy (TEM). and found the gold initially forms new clusters, but at "a certain size, no new agglomerates are formed, but existing agglomerates continue to increase in size" [82]. Similarly, Robinson, using TEM showed a difference in nanocluster size and density when gold was deposited into stearic acid versus a carbon thin film, but the difference all but disappeared when the stearic acid was treated with a glow discharge [83]. The use of glow discharge pre-treatment by Robinson [83,84] as well as Holland, Johnson, and Thomas and Kent [30,31,85] as a way to remove contaminants or to sensitize the substrate surface was never adapted for operational use, but is of particular interest to this paper and will be further discussed below.

3. Current literature theories

The theories of the fVMD deposition mechanism currently found in the literature are summarized below. Before going further, the reader should be aware of a few general concepts. First, all were all derived from an initial examination of gold-zinc deposition. Any examination of mono-metal deposition was secondary or via extrapolation. Second, the deposition of zinc. Most of the focus has on the deposition of gold as the critical step. The mechanism of zinc deposition has also been presumed via extrapolation. In each case, the presumption is the zinc fails to deposit (either via a failure to adsorb or desorption) unless a gold seed is present on the surface to act as a nucleation site. This, however, has not been fully tested. With these limitations in mind, the three current literature theories are presented below.

3.1. The condensation theory

First proposed by Stroud [33,34], the condensation theory is based on classical nucleation theory and states the different condensation and surface diffusion energies of a substrate will result in a differential condensation/desorption rate between the different substrates (see Fig. 4). Derived from Equation 1, where the critical condensation condition (Qcrit), and holding N0 (number of nucleation sites), V0 (fundamental vibrational frequency), k (Boltzmann constant) and T (temperature in K) as constant, a shift in the condensation energy (Ea) from 0.2 eV to 0.3 eV after 10 s results in a 1000-fold shift in cluster density between the two [34]. Stated otherwise, the calculations purport to show that, due to desorption, there would be significantly fewer gold clusters on the matrix vs the substrate.

$$Q_{crit} \geq N_0 V_0 \left(-\frac{E_a + E_b}{kT}\right)$$

As mentioned earlier however, this theory was found to be lacking by both Collins *et al.,* via neutron activation analysis [26] and Jones *et al.*, via ICP-MS [79,80] who showed that the quantity of gold deposited was consistent over the analyzed area, independent of the presence of a fingerprint. This idea continues to be cited however, with Steiner and Bécue stating desorption and reflection as a function of surface energy being the main parameter behind cluster formation [14].

3.2. The absorption theory

First proposed by Collins et al. [26], the absorption theory is based on the idea the gold nanoparticles adsorb onto the surface, but are then absorbed into the sebaceous material of the matrix and are therefore no longer on present on the surface to act as nucleation sites for the subsequent zinc deposition (see Fig. 5 for a summary). Specifically, from Collins et al., pg. 16 [26], "It may be that during the gold evaporation the deposited gold becomes intermixed with the grease and print remains essentially as a surface grease pattern, which inhibits the

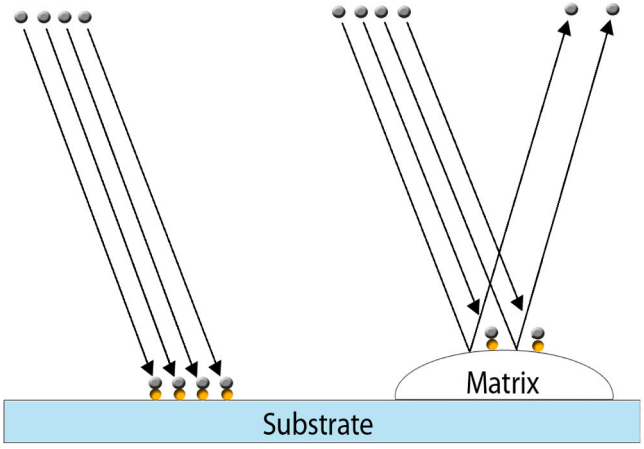


Fig. 4. – Visualization of Stroud's condensation theory. A difference in condensation energy results in fewer gold atoms depositing on the surface of the matrix, which results in fewer binding sites for the subsequent zinc deposition. The result is fewer zinc atoms binding on the matrix.

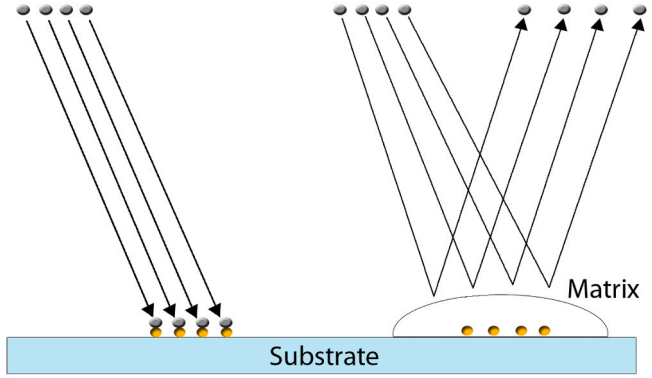


Fig. 5. – Visualization of Collin's absorption theory. The gold atoms absorb into the matrix and are therefore not present on the surface to act as binding sites for the subsequent zinc deposition.

subsequent deposition of cadmium."

This is currently the *de facto* theory [1,4]. However, there is no definitive proof for this as the initial experiments were observational or using indirect measurements, in part due to the limitations of the technology at the time.

3.3. The morphology theory

First proposed by Jones [80], the morphology theory builds on the idea that zinc will only nucleate onto a specific size or shape of gold nanoparticle or smaller. After that specific size, termed the critical morphology, the gold nanoparticles will no longer be able act as nucleation sites for the zinc (see Fig. 6 below). Specifically, the gold absorbs into the sebaceous material of the matrix while growing normally on the substrate. After sufficient deposition, the nanoparticles on the matrix reach the surface and are once again able to act as nucleation sites for the zinc. By that point, the gold nanoparticles on the substrate may have surpassed the critical morphology, preventing subsequent zinc deposition and thereby creating the situation for reverse development.

However, per Jones, an additional, substrate-specific effect was observed whereby the growth occurred differently on different substrates. A summarized explanation and figures from Jones [80] for this effect is shown below as described in relation to the differential deposition between LDPE and HDPE plastics:

- For LDPE plastics (see Fig. 7 below).

o Low levels of gold deposition: The deposited gold forms clusters. On the substrate, these form on the surface. On the matrix, these clusters are absorbed into the sebaceous material. At this point, if zinc is deposited, the zinc cannot bind to the gold in the matrix, and it results in standard development.

o Medium levels of gold deposition: As further gold is deposited, the clusters both in the matrix and outside of the matrix continue to grow. At a certain point, the clusters inside the matrix grow to the point where they emerge from the matrix, becoming able to act as nucleation sites for the zinc. Meanwhile, the clusters on the substrate have surpassed the critical morphology, meaning they can no longer act as nucleation sites for the zinc, resulting in reverse development.

o High levels of gold deposition: As even more gold is deposited, the clusters both in the matrix and on the substrate have surpassed the critical morphology, rendering both incapable of acting as nucleation sites for the zinc.

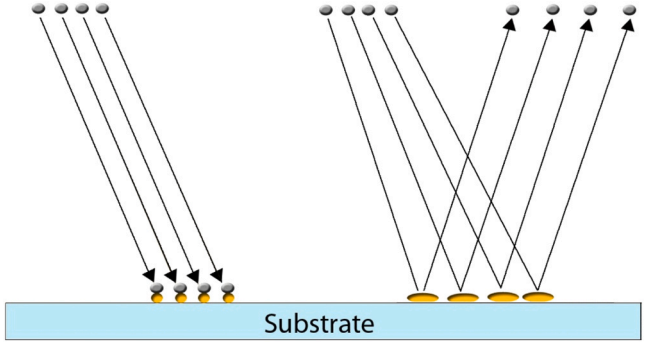


Fig. 6. – General visualization of Jone's morphology theory. The gold nanoparticles grow and continue to act as nucleation sites for the zinc until the critical morphology is reached. Once the critical morphology is reached, the zinc atoms are no longer able to deposit.

For HDPE plastics (see Fig. 8 below).

o Low levels of gold deposition: The deposited gold forms clusters. On the substrate, these form on the surface. On the matrix, these clusters are absorbed into the sebaceous material. At this point, if zinc is deposited, the zinc cannot bind to the gold in the matrix, and it results in standard development.

o Medium levels of gold deposition: As further gold is deposited, the clusters both in the matrix and outside of the matrix continue to grow. At a certain point, the clusters inside the matrix grow to the point where they emerge from the matrix, becoming able to act as nucleation sites for the zinc. Meanwhile, the clusters on the substrate have not yet surpassed the critical morphology, meaning they can still act as nucleation sites for the zinc, resulting in no visible contrast being developed (over-development).

o High levels of gold deposition: As even more gold is deposited, the clusters in the matrix have surpassed the critical morphology, rendering it incapable of acting as nucleation sites for the zinc. Meanwhile the clusters on the substrate have not reached the critical morphology, resulting in normal, but poor-quality development.

Building on the absorption theory, it still depends on the absorption of the nuclei into sebaceous material. However, the addition of the morphological component offers a clearer explanation for aspects such as reverse development.

This completes the evaluation of the literature on fVMD. A summarization of the current standing of fVMD theory follows below.

3.4. Current understanding of fVMD summarized

Based on the information gathered so far, the reader can see several points of consensus as well as a few points of dissension. For the points of consensus, fVMD:

1) Is a highly sensitive technique [1,4,5,7,8,11,12,16,18–22,24–27,29,36,38,40–42,44–47,53,54,56,58–60,67,68,77,79–81,86–88].
2) Is highly dependent on the substrate [1,3–5,7,9,11–13,21,29,36,40,41,47,58,60,62,70,74, 76,79–81,86,89]
3) Functions primarily due to the presence sebaceous components of the matrix [1,4,5,7,9,10,14,15,19,21,29,33,34,36,38,50,51,56,58,60,62,67–71,77,80,81,89,90]
4) Works on difficult exhibits including, but not limited to:

a. plastics, including polyethylene [1,4,5,13,15,29,40,41,47,50,73,80,81,86]
b. ballistic material [18–20, 62, 70]
c. banknotes [11–13, 55, 76, 77]
d. aged impressions [1, 3–5, 7, 8, 21, 23, 45]
e. wetted impressions [3,14,21,25,53,55]

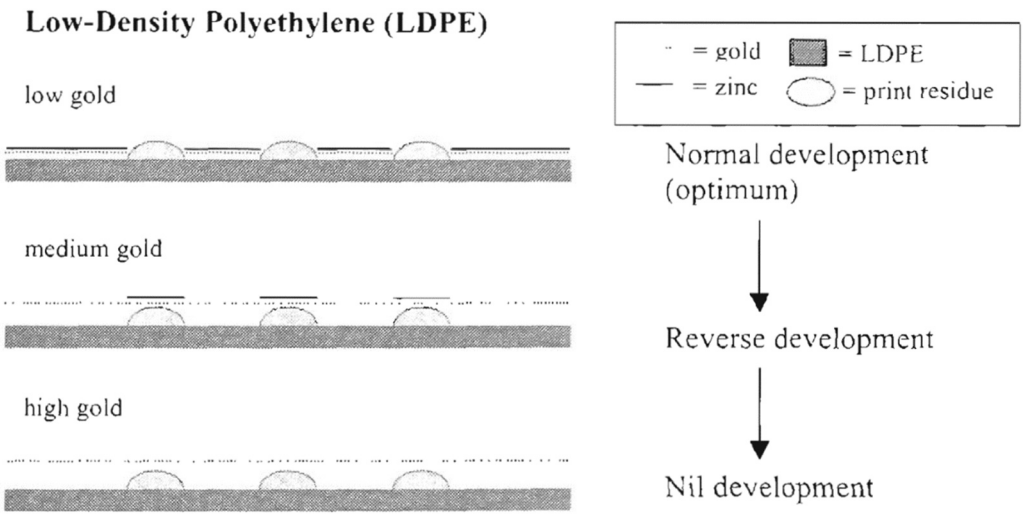


Fig. 7. – Gold-zinc development on LDPE, depending on the amount of gold deposited initially. The mechanism of fVMD development on LDPE, as per the morphology theory. The full explanation continues below. Image source Jones [79], pg. 46.

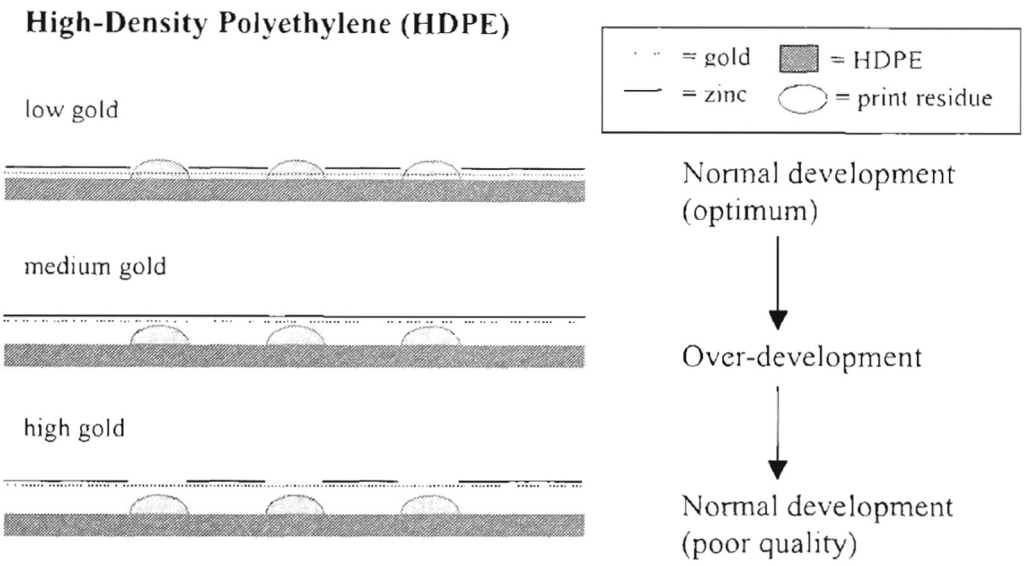


Fig. 8. – Gold-zinc development on HDPE, depending on the amount of gold deposited initially. The mechanism of fVMD development on LDPE, as per the morphology theory. The full explanation continues below. Image source Jones [79], pg. 47.

5) Can be done with a variety of metals [1,5,14,21,24–26,37,38,51,61, 65–68,77,89,90]
6) Often uses AuZn as the default [1, 3–5, 7, 9, 14, 15, 29, 36, 40, 41, 45, 48, 50, 58, 60, 62–64, 66, 70, 80–82, 86, 89]
7) When performing AuZn deposition, the gold acts as nucleation sites seeding the subsequent zinc deposition [1, 3–5, 7, 9, 36, 41, 58, 60, 62, 70, 80, 81, 86, 89]

The consensus regarding the operational utility of the technique is therefore not in question. But the mechanics of the deposition remains shrouded. As previously mentioned, three theories have been posited and are described below in more detail.

4. Characteristics of fVMD – remaining questions

The current consensus, as per the *Fingermark Visualisation Manual*, is the gold nuclei absorb into the grease, resulting in them no longer being on the surface to act as nucleation sites for the zinc (the absorption theory). It does however note, in a nod to the morphology theory, that "there is no conclusive evidence of nuclei diffusion and it is possible that the effect observed may be solely attributable to different growth rates of zinc on regions where gold nuclei of different sizes have been formed." [1]

However, the absorption of the gold is not the only point of ambiguity. Based on this literature review, there remains numerous details that cannot be explained by the current theories. This includes:

4.1. Mono-metal deposition

[1,5,14,21,24–26,37,38,51,61,65–67,77,89]. There is currently no mechanism posited which can explain the colour shift between the matrix and the substrate when performing mono-metal deposition, by any of the three theories. The closest are statements by Philipson and Bleay who attribute the color difference to differences in nuclei size and density altering the colors of the transmitted light [67]; Steiner and Bécue, who attribute the colour shift to variations in the Rayleigh and Mie scattering [14] and Collins, who states "the colour of gold films depends on the crystal size (of the gold)" [26]. In each case, the

explanations were offered, but with no cited data or theoretical calculations.

4.2. Empty prints

[9,14,48,60,61,67,79–81]. The current explanation comes from Jones [13,80,81] and involves the matrix seeping into the furrows. This was based on the observation that empty prints were more frequent with fresh and heavily loaded impressions on PE and PVC. While the results from Popov, Sears and Jones seems to offer some support for this theory [71], Philipson and Bleay showed subsequent treatment using Ag shows no generalized diffusion of the matrix into the furrows [67], and Steiner and Bécue showed empty prints occurring on even natural, aged impressions [14]. As the explanation for the empty prints is simply an expansion of the matrix, all three theories suffer the same limitations as when applied to normal depositions.

4.3. Reverse development

[1,13,21,29,40,48–50,74,80,81]. The current explanation involves the organic residue of the matrix absorbing into the polymer, leaving only solid, inorganic residues on the surface into which the gold nuclei cannot penetrate [1,40]. However, Jones found the incidence of reverse development was dependent on the amount of gold deposited, at least for LDPE [79,80]. Grant, Springer and Ziv found the normal development for younger prints, but reverse development on aged prints [50]. Masters observed reverse development impressions deposited onto soda-lime glass [48]. Johnson was able to create reverse development on fingerprints on glass and cotton following glow discharge treatment [32]. Smith noted reverse development and attributed it to the level of sodium chloride crystals [49]. Further, as noted above, Jones observed not just reverse development, but reverse empty prints on PP substrates, noting specifically "The (PP) results do not fit with the present understanding of fVMD development" [81]. Also of note is the observation that reverse development also tends to occur following cyanoacrylate fuming [36]. Of the three theories the morphology theory best explains this effect, but it does not appear to fully explain the effect. The condensation theory could explain it, requiring only an inversion of the condensation energies relative to a normal deposition scenario.

4.4. Development on heated prints [16,17]

. Even after exposure to elevated temperatures for extended periods of time, periods after which all sebaceous material would have long since combusted or evaporated, fVMD was still able to develop usable prints. The ability to show development where the sebaceous materials cannot remain shows yet another exception to the absorption theory. As both the absorption and morphology theory require a permeable organic layer, neither can explain this phenomenon. The condensation theory could potentially apply as the condensation energy of a surface is not necessarily dependant on the presence of matrix.

4.5. Impact of glow discharge treatment [30,31,83–85]

Pretreatment of a surface using glow discharge has a drastic effect on subsequent fVMD treatment. But as glow discharge only affects the surface and not the underlying sebaceous material, absorbed nanoparticles should theoretically be unaffected.

Therefore, despite the consensus elevating the absorption theory of fVMD development, it is clear the theory is, at best, lacking. All three are dependent on an active process dependent on effects that occur during the deposition step. The condensation theory appears to be the best suited to apply across all scenarios but suffers from two independent experiments showing there is no evidence of any variation in the quantity of gold being deposited. None of the theories have stated a mechanism for the colour change in active deposition. None of the theories can explain the effect of empty prints, albeit limited by a lack of understanding of the mechanism by which empty prints form. The morphology or condensation theories can potentially explain reverse development. Only the condensation theory can explain the development on impressions where the matrix had been removed by heating. Therefore, none of the theories, as they are currently presented, can explain all the observed phenomena. Further research is therefore required.

If we can understand the mechanism, we can better understand the technique. A better understanding of the technique could lead to more a better targeting of the fVMD. At the moment, the primary limitation for the fVMD is one of resources [7]. A better understanding could lead to a more effective or targeted treatment process, potentially allowing for more and better fingerprints developed, thus better serving the cause of justice. If sebaceous materials are truly not required, this opens whole new avenues of forensic treatment: arson exhibits; eccrine prints; toolmarks or firearm marks; questioned document examination. At the very least, better understanding the mechanism of fVMD will give more credibility to the field in the courts, allowing an expert witness to avoid the response of "we don't really know", when asked by a court to explain how the fVMD works.

5. Conclusion

This paper summarizes the current understanding of forensic vacuum metal deposition (fVMD), it's operational use and the proposed mechanisms of development. It shows fVMD has been in use for over 50 years and is a fully integrated technique that is renowned for its sensitivity and utility in operational work, particularly on aged impressions, wetted impressions and difficult substrates. However, despite the years of research and operational use, the precise mechanism of development remains unclear. There are three posited theories: the absorption theory, the condensation theory and the morphology theory. Of the three, the morphology theory seems to best explain the observed effects. However, there is no conclusive proof any of the theories are correct. Furthermore, this paper demonstrates several observed scenarios which the current theories seem unable to fully explain. It can therefore be concluded the current theories are, at best, incomplete, and that further research is required.

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CRediT authorship contribution statement

Andreas Ruëdiger: Writing – review & editing, Supervision, Project administration, Funding acquisition. **Benoit Daoust:** Writing – review & editing, Supervision. **Aaron Dove:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization.

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References

[1] H.L. Bandey. Fingermark Visualisation Manual, 2nd edition, Defense Science and Technology Laboratory, 2022.
[2] A. Gaw, Lee and Gaensslen's Advances in Fingerprint Technology, CRC Press, 2012, https://doi.org/10.1201/b12882.
[3] G.W. Batey, J. Copeland, C.L. Hill, P.L. Laturnus, C.H. McDiarmid, K.J. Miller, A. H. Misner, A. Tario, A.B. Yamashite, Metal Deposition for Latent Print Development, J. Forensic Identif. 48 (2) (1998) 165–175.
[4] E. Holder, The fingerprint sourcebook; U.S. Dept. of Justice, Office of Justice Programs, National Institute of Justice Washington, DC, 2011.
[5] A. Exall, Fingermark Visualisation Source Book, 3, Home Office Defense Science and Technology Laboratory, 2022.
[6] Kahn, R. What Really Happened With the 'Last Call' Killer Who Terrorized NYC's Gay Nightspots in 1980s and '90s? A&E, 2023. ⟨https://www.aetv.com/real-crime/last-call-killer⟩ (accessed 2023/11/06).
[7] A. Dove, The use of vacuum metal deposition in operational casework: a 2 years retrospective, Identif. Can. 40 (2017) 112–127.
[8] Dove, A. Case report: A 41-year-old fingerprint in a cold case using VMD, 105th Internation Association for Identification Annual Conference, Nashville, 2021.
[9] J. Fraser, P. Deacon, S. Bleay, D.H. Bremner, A comparison of the use of vacuum metal deposition versus cyanoacrylate fuming for visualisation of fingermarks and grab impressions on fabrics, Sci. Justice 54 (2) (2014) 133–140.
[10] A. Horvath, Visualization of latent fingerprints on fabrics using vacuum metal deposition, J. Forensic Identif. 72 (3) (2022) 339–370.
[11] R. Lam, D. Wilkinson, T. Tse, B. Pynn, Recommended protocols for fingerprint detection on canadian polymer banknotes - Part 1: chemical development, J. Forensic Identif. 64 (2014) 375–401.
[12] J. Flynn, M. Stoilovic, C. Lennard, Detection and enhancement of latent fingerprints on polymer banknotes: a preliminary study, J. Forensic Identif. 49 (6) (1999) 594–613.
[13] N. Jones, M. Kelly, M. Stoilovic, C. Lennard, C. Roux, The development of latent fingerprints on polymer banknotes, J. Forensic Identif. 53 (1) (2003) 50–77.
[14] R. Steiner, A. Becue, Effect of water immersion on multi- and mono-metallic VMD, Forensic Sci. Int. 283 (2018) 118–127.
[15] Z. Ziv, E. Springer, Additional Methods for Pre-Treatment of Drug Contaminated Polythene Bags Prior to Vacuum Metal Deposition of Latent Fingerprints. International Symposium on Fingerprint Detection and Identification, Neu'rim, June 26-30, 1995, National Police, Israel, 1995, pp. 179–196.
[16] A.J. Dominick, N. Nic Daeid, S. Bleay, The recoverability of fingerprints on nonporous surfaces exposed to elevated temperatures, J. Forensic Identif. 61 (5) (2010) 520–536.
[17] S. Bleay, G.W. Bradshaw, J.E. Moore, Fingerprint Development and Imaging Newsletter: Special Edition, Home Office Scientific Development Branch, Sandridge, 2006.
[18] E. Brewer, The capability of forensic vacuum metal deposition for developing latent fingermarks on fired ammunication - a preliminary study comparing alternative metals processes, J. Forensic Identif. 69 (2018) 299–327.
[19] E. Brewer, Visualisation of latent fingermark detail on fired handgun casings using forensic VMD, J. Forensic Identif. 70 (2020) 323–345.
[20] J. Shipman, Evaluation of vacuum metal development on plastic gloves, handguns and live ammo, J. Forensic Identif. 71 (2021) 119–141.
[21] D.S. Hambley, University of London, Phys. Vac. Evaporation Dev. Latent Fingerpr. (1972).
[22] Tolansky, D.S. Home Office Scientific Advisory Branch, Ed., 1964.
[23] P. Theys, Y. Turgis, G. Lepareux, G. Chevet, P.F. Ceccaldi, New technique for bringing out Latent Fingerprints on Paper: Vacuum Metallization, Int. Crim. Police Rev. 217 (1968) 106–108.
[24] Coles, R.E.; Collins, L.E. The Development of Latent Fingerprints on Cloth Progress Report March 1973 - March 1974; AWRE, 1974.
[25] Coles, R.E.; Collins, L.E. The Development of Latent Fingerprints on Cloth Progress Report April 1974 - March 1975; AWRE, 1975.
[26] L.E. Collins, R.E. Coles, P.T. Stroud. The Development of Fingerprints on Cloths Progress Report March 1972 - February 1973, AWRE, 1973.
[27] G.L. Thomas, T.E. Reynoldson, Some observations on fingerprint deposits, J. Phys. D Appl. Phys. 8 (1975) 724–729.
[28] G.L. Thomas, A fingerprint thin film, Thin Solid Films 24 (1974) S52–S54.
[29] T. Kent, G.L. Thomas, T.E. Reynoldson, H.W. East, A vacuum coating technique for the development of latent fingerprints on polythene, J. Forensic Sci. Soc. 16 (1976) 93–101.
[30] G.L. Thomas, K. Kent, On the Use of CF Glow Discharge to Remove Contamination Prior to Fingerprint Development by Metal Deposition, Home Office, 1975.
[31] R. Johnson, G.L. Thomas, T. Kent, Etching of fatty acid films in a glow discharge, Thin Solid Films 38 (1976) L13–L15 (Letter).
[32] R. Johnson, The Use of a Glow Discharge in the Development of Fingerprints by Metal Evaporation, Home Office (1976).
[33] P.T. Stroud, Further comments on finger print development by vacuum deposition, AWRE (1972).
[34] Stroud, P.T. Some Comments on Finger Print Development by Vacuum Deposition.; 5/71; AWRE, 1971.
[35] R. Downham, S. Mehmet, V.G. Sears, A Pseudo-operational investigation into the development of latent fingerprints on flexible plastic packaging films, J. Forensic Identif. 62 (6) (2012) 661–682.
[36] B.J. Jones, R. Downham, V.G. Sears, Nanoscale analysis of the interaction between cyanoacrylate and vacuum metal deposition in the development of latent fingermarks on low-density polyethylene, J. Forensic Sci. 57 (1) (2012) 196–200.
[37] T.A. Trozzi, R.L. Schwartz, M.L. Hollars, Processing Guide for Developing Latent Prints, US. Department of Justice Federal Bureau of Investigation, 2000.
[38] G. Evans, A. Horvath, Forensic VMD Application Guide (Ed), in: Price, J, 1, West Technology Forensics, 2020.
[39] P. Margot, C. Lennard, Fingerprint Detection Techniques, Université de Lausanne, 1994.
[40] Kent, T.; Thomas, G.L.; East, H.W. Application of the Metal Deposition Technique to the Development of Fingerprints on Polythene; London, 1975.
[41] T. Kent, User Guide to the Metal Deposition Process for the Development of Latent Fingerprints, 24/82, Home Office, 1982.
[42] C.A. Pounds, R.J. Jones, D.G. Sanger, The use of Powder Suspensions for Developing Latent Fingerprints. Part 2. Assessment on Paper, Polythene and Window Glass Surfaces, 5, Home Office Central Research Establishmend, 1981.
[43] D.S. Smith, The Development of Latent Fingeprints on PVC, South Bank University, 1981.
[44] Reynoldson, T.E.; Reed, F.A. Operational Trial Comparing Metal Deposition with Small Particle Reagent for the Development of Latent Prints on Polyethylene; London, 1979.
[45] A.H. Misner, Latent fingerprint detection on low density polyethylene comparing vacuum metal deposition to cyanoacrylate fuming and fluorescence, J. Forensic Identif. 42 (1) (1992) 26–48.
[46] A.H. Misner, TM-15-93 Fingerprint Detection Using Vacuum Metal Deposition, Canadian Police Research Centre, 1993.
[47] T. Kent, P.C. Gillet, D. Lee, A Comparative Study of Three Techniques; Aluminium Powdering, Lead Powdering and Metal Deposition for the Development of Latent Fingerprints on Polythene, Home Office, London, 1978.
[48] N.E. Masters, J.D. DeHaan, Vacuum metal deposition and cyanoacrylate detection of older latent prints, J. Forensic Identif. 46 (1996) 32–45.
[49] Smith, G. Thin film deposition for fingerprint detection in criminal investigations. In 32nd Annual Technical Conference of the Society of Vacuum Coaters, Missouri, 1989; pp 320-324.
[50] H. Grant, E. Springer, Z. Ziv, Vacuum Metal Deposition Inhibition on Polythene Bags, June 26-30, 1995. International Symposium on Fingerprint Detection and Identification, Neu'rim, Israel National Police, 1995, pp. 203–215. June 26-30, 1995.
[51] R.N. Morris, C.L. Gilmour, P.L. Latumus, A.B. Yamashita, Vacuum metal deposition of silver as an aid in credit card examinations, J. Forensic Identif. 47 (1) (1997) 57–63.
[52] J. Meikle, K. Jones, S.L. Cresswell, C. Matheson, J.F. Carter, The effects of fingerprinting agents on the stable isotope ratios of polyethylene films, Sci. Justice 64 (6) (2024) 599–604.
[53] J.W. Godsell, P.G. Vincent, D.W. Lloyd, 1972, Influence of storage of latent fingerprints in an atmosphere of a given humidity on the choice of the optimum developing agent, 3/72.
[54] P.B. Sherrifs, K.J. Farrugia, J.M. Fraser, B.J. Jones, The optimisation of fingermark enhancement by VMD and Lumicyano™ on thermal paper, Sci. Justice 60 (2) (2020) 160–168.
[55] C. Wang, Z. Qian, W. Li, Y. Luo, Comparison of vacuum metal deposition and 1,2-indandione/ninhydrin reagent method for the development of fingerprints on Renminbi, J. Forensic Sci. Med. 3 (2) (2017).
[56] S. Suzuki, Y. Suzuki, H. Ohta, Detection of latent fingerprints on newly developed substances using the vacuum metal deposition method, J. Forensic Identif. 52 (5) (2002) 573–578.
[57] R. Albinson, The Development of Fingerprints on Fabrics - Laboratory Simulations Using a Range of Development Techniques. E.E.C International Fingeprint Conference, Home Office Scientific Research and Development Branch, London, 1984.
[58] J. Fraser, A comparison of the use of vacuummetal deposition versus cyanoacrylate fuming for visualisation of fingermarks and grab impressions on fabrics, Sci. Justice 54 (2014) 133–140.
[59] J. Fraser, Abertay University, Enhanced Tech. Fingermark Recov. Fabr. (2013).
[60] J. Fraser, K. Sturrock, P. Deacon, S. Bleay, D.H. Bremner, Visualisation of fingermarks and grab impressions on fabrics. Part 1: gold/zinc vacuum metal deposition, Forensic Sci. Int. 208 (1-3) (2011) 74–78 (From NLM PubMed-not-MEDLINE).
[61] S. Knighting, J. Fraser, K. Sturrock, P. Deacon, S. Bleay, D.H. Bremner, Visualisation of fingermarks and grab impressions on dark fabrics using silver vacuum metal deposition, Sci. Justice 53 (2013) 309–314.
[62] G. Christofidis, J. Morrissey, J.W. Birkett, A preliminary study on vacuum metal deposition as a standalone method for enhancement of fingermarks on ballistic brass materials, J. Forensic Sci. 64 (5) (2019) 1500–1505.
[63] A.L. Chelli, Testing the efficacy of experimental metals in the vacuum metal deposition process for the development of latent fingerprints, California State University Long Beach, 2007.
[64] A. Gunarante, C. Knaggs, D. Stansbury, Vacuum metal deposition: comparing conventional gold/zinc VMD to aluminum VMD, Identif. Can. (2007).
[65] I.H. Yu, S. Jou, C.-M. Chen, K.-C. Wang, L.-J. Pang, J.S. Liao, Development of latent fingerprint by ZnO deposition, Forensic Sci. Int. 207 (1) (2011) 14–18.

[66] A.K. Tiwari, I.M. Alaoui, S. Guddala, S.A. Ramakrishna, Enhanced visualization of latent fingermarks on rough aluminum surfaces using sequential Au and Zn/ZnS/ZnO depositions, J. Forensic Sci. 63 (4) (2018) 1275–1281.
[67] D. Philipson, S. Bleay, Alternative metal processes for vacuum metal deposition, J. Forensic Identif. 57 (2) (2007) 252–273.
[68] C. Gaskell, S.M. Bleay, H. Willson, S. Park, The enhancement of fingermarks on grease-contaminated, nonporous surfaces: a comparative assessment of processes for light and dark surfaces, J. Forensic Identif. 63 (2013) 286–319.
[69] N. Nic Daeid, S. Carter, K. Laing, A comparison of vacuum metal deposition and powder suspension for recovery of fingerprints on wetted nonporous surfaces, J. Forensic Identif. 58 (5) (2003) 600–613.
[70] J.N. Pollitt, G. Christofidis, J. Morrisey, J.W. Birkett, Vacuum metal deposition enhancement of friction ridge detail on ballistic materials, Forensic Sci. Int. 316 (2020).
[71] K.T. Popov, V.G. Sears, B.J. Jones, Migration of latent fingermarks on non-porous surfaces: Observation technique and nanoscale variations, Forensic Sci. Int. 275 (2017) 44–56.
[72] N.J. Bright, T.R. Willson, D.J. Driscoll, S.M. Reddy, R.P. Webb, S. Bleay, N.I. Ward, K.J. Kirkby, M.J. Bailey, Chemical changes exhibited by latent fingerprints after exposure to vacuum conditions, Forensic Sci. Int. 230 (1–3) (2013) 81–86.
[73] P.F. Edwards, The Use of Sputtered Thin Films in the Development of Latent Fingerprints on Glass and Polythene, 43/76, Home Office Police Scientific Development Branch, 1976.
[74] K. Kent, M. Stoilovic, Development of latent fingerprints using preferential DC sputter deposition, Forensic Sci. Int. 72 (1995) 35–42.
[75] I.J. Turner, M.A. Burgess, J.M. Love, P.T. Lynch, Sputter coating a viable alternative for the development of latent prints on non-porous surfaces, Forensic Sci. J. 9 (1) (2010) 19–24.
[76] R. Newton, South Bank University, Detect. latent Fingerpr. Banknotes Met. Depos. (1975).
[77] L.W.L. Davis, P.F. Kelly, R.S.P. King, S. Bleay, Visualisation of latent fingermarks on polymer banknotes using copper vacuum metal deposition: a preliminary study, Forensic Sci. Int. (2016).
[78] R. Lam, The use of vacuum metal deposition for forensic casework in Canada, Aust. J. Forensic Sci. 51 (sup1) (2019) S124–S127.
[79] N. Jones, Metal Deposition Techniques for the Detection and Enhancement of Latent Fingerprints on Semi-Porous Surfaces, University of Technology, Sydney, NSW, Australia, 2002.
[80] N. Jones, M. Stoilovic, C. Lennard, C. Roux, Vacuum metal deposition: factors affecting normal and reverse development of latent fingerprints on polyethylene substrates, Forensic Sci. Int. 115 (2001) 73–88.
[81] N. Jones, D. Mansour, M. Stoilovic, C. Lennard, C. Roux, The influence of polymer type, print donor and age on the quality of fingerprints developed on plastic substrates using vacuum metal deposition, Forensic Sci. Int. 124 (2001) 167–177.
[82] X. Dai, M. Stoilovic, C. Lennard, N. Speers, Vacuum metal deposition: visualisation of gold agglomerates using TEM imaging, Forensic Sci. Int. 168 (2–3) (2007) 219–222.
[83] C.J. Robinson, The effects of a glow discharge on the nucleation characteristics of gold on polymer substrates, Thin Solid Films 57 (2) (1979) 285–289.
[84] C.J. Robinson, The effects of glow dischage surface treatment on the nucleation of thin solid films on organic substrates, Univ. Sussex (1978).
[85] L. Holland, G.L. Thomas, Progress Report on Glow Discharge Treatment of Fingerprints for the period 1st October 1976–1st April 1977, University of Sussex, 1977.
[86] T. Kent, Latent fingerprints and their detection, J. Forensic Sci. Soc. 21 (1) (1981) 15–22.
[87] G.L. Thomas, Physical methods of fingerprint development, Can. Soc. Forensic Sci. J. 8 (4) (1975) 144–154.
[88] L. Woodcock, N. Coogan, J. Ringe, N. Frascione, Enhancing fingermark visualisation on pangolin scales and rhino horns using cyanoacrylate fuming and vacuum metal deposition techniques, Forensic Sci. Int. 376 (2025) 112584.
[89] H. Dafydd, G. Williams, S. Bleay, Latent fingerprint visualization using a scanning Kelvin probe in conjunction with vacuum metal deposition, J. Forensic Sci. 59 (1) (2014) 211–218.
[90] D. Charlton, A. Dove, S. Hinder, C. Costa, A. Ruëdiger, J.F. Watts, M.J. Bailey, Sequential application of Time of Flight-Secondary Ion Mass Spectrometry after vacuum metal deposition on glass, polyethylene terephthalate and polyvinyl chloride, Science & Justice 65 (6) (2025).